

Scope Boundary Validation — Wave W004

Phase 8

Project: MINERVA CryoCell / QCELL & RFCELL P&ID

Wave: W004 (Geometric Tracing), Phase 8

1. Summary

Metric	Count
Scope boundaries detected	5 / 5
Handover diamonds (TP#NNN)	22
W-line bottom-right elements isolated	19
Ambiguities	0

All five expected scope boundaries were located by text-matching against the title block and annotation layer. Handover points use the `TP#NNN` pattern (TP#101 ... TP#604) rather than the generic `TPXXXX` placeholder; the parser regex `\bTP[#A-Z]?\d{2,5}\b` matches both.

2. Boundaries detected

Boundary	Evidence (sample)
QM (Cryomodule)	"Cryomodule (QM)", "QCELL-QM", "scope QM"
QVB	"scope QVB", "scope QVB (AUB)", "QVB INVAC connection CF40", "vacuum barrier"
vacuum_barrier	"vacuum barrier"
Jumper	"Jumper"
QINFRA	"QINFRA - Implementation by NFS", "scope QINFRA"

3. Handover diamonds (22)

Matched on the `TP#NNN` pattern. Each carries sheet + text + resolved (x, y) in viewBox coordinates. Series observed:

- **1xx series:** TP#101, TP#102, TP#103, TP#104, TP#105
- **4xx series:** TP#401, TP#402, ... (QCELL lower band)
- **6xx series:** up to TP#604

Full list with coordinates in

`data/model/scope_boundary_validation.json` → `handover_diamonds_TPXYYYY`.

4. W-line bottom-right zone

19 elements belonging to the W (coupler, GREEN) line were isolated in the bottom-right quadrant of the drawing for visual review (atlas layer `04C_Lines_W_GREEN`). Quadrant math uses the parsed **viewBox bounds** (default `[0,0,1527.27,1080]`) — not raw min/max of coordinates — to avoid a known degenerate-transform coordinate outlier.

5. Continuity flags

1. W line bottom-right zone isolated for visual review (see atlas layer 04C).
2. Handover diamonds matched on TPXYYYY pattern; verify category prefix per AD_01.16.

6. Provenance

- Engine: `src/abacus_svg_pid/build_w003_w004.py` (Phase 8)
- Machine-readable output: `data/model/scope_boundary_validation.json`